

Algebra 1
Unit 5
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

Solve each system of equations using substitution for questions 1-4.

$$1. \begin{cases} 3x + y = -9 \\ x = 6y \end{cases} \quad \begin{aligned} 3(6y) + y &= -9 \\ 18y + y &= -9 \\ 19y &= -9 \\ y &= -\frac{9}{19} \end{aligned} \quad \begin{aligned} x &= 6\left(-\frac{9}{19}\right) \\ x &= -\frac{54}{19} \end{aligned}$$

$\left(-\frac{54}{19}, -\frac{9}{19}\right)$

$$2. \begin{cases} y = 2x \\ y = -2x + 12 \end{cases} \quad \begin{aligned} 2x &= -2x + 12 \\ 4x &= 12 \\ x &= 3 \end{aligned} \quad \begin{aligned} y &= 2(3) \\ y &= 6 \end{aligned}$$

$(3, 6)$

$$3. \begin{cases} 2x + 3y = 5 \\ x = -2y \end{cases} \quad \begin{aligned} 2(-2y) + 3y &= 5 \\ -4y + 3y &= 5 \\ -y &= 5 \\ y &= -5 \end{aligned} \quad \begin{aligned} x &= -2(-5) \\ x &= 10 \end{aligned}$$

$(10, -5)$

$$4. \begin{cases} 4y - 1 = x \\ 8x + 7y = -47 \end{cases} \quad \begin{aligned} 8(4y - 1) + 7y &= -47 \\ 32y - 8 + 7y &= -47 \\ 39y &= -39 \\ y &= -1 \end{aligned} \quad \begin{aligned} 4(-1) - 1 &= x \\ -5 &= x \end{aligned}$$

$(-5, -1)$

Use the system $\begin{cases} 3x - 2y = 11 \\ y = \frac{1}{2}x - 1 \end{cases}$ for questions 5 - 6.

5. Describe why using substitution is an efficient method to solving the system.
Substitution is an efficient method to solve a system of equations when one variable is already isolated or solved for in terms of the other variable.

6. Solve the system.

$$\begin{aligned} 3x - 2\left(\frac{1}{2}x - 1\right) &= 11 & y &= \frac{1}{2}\left(\frac{9}{2}\right) - 1 \\ 3x - x + 2 &= 11 & y &= \frac{9}{4} - \frac{4}{4} \\ 2x &= 9 & y &= \frac{5}{4} \\ x &= \frac{9}{2} \\ \left(\frac{9}{2}, \frac{5}{4}\right) &\text{ or } (4.5, 1.25) \end{aligned}$$

John and Max work at a sandwich shop. John can make 15 sandwiches per hour, and Max can make 10 sandwiches per hour. Max worked 5 more hours than John and they made a total of 150 sandwiches that day. Determine the number of hours Max worked and the number of hours John worked.

7. Name the *two* variables that are being related in this situation.

Let m represent the number of hours Max works.

Let j represent the number of hours John works.

8. Write a system of equations to represent the number of hours Max and John worked.

$$\begin{cases} 15j + 10m = 150 \\ m = j + 5 \end{cases}$$

9. Solve the system of equations using substitution.

$$15j + 10(j + 5) = 150$$

$$m = 4 + 5$$

$$15j + 10j + 50 = 150$$

$$m = 9$$

$$25j = 100$$

$$j = 4$$

10. Interpret the solution of the solution.

John worked for 4 hours, and Max worked for 9 hours to make a total of 150 sandwiches.